

The **MDR** (an acronym for **micro dynamic rifle**) is a family of autoloading bullpup rifles designed by Desert Tech (formerly Desert Tactical Arms) in 2014. A second generation version of the rifle is marketed as the **MDRX** (an acronym for **micro dynamic rifle eXtreme**).^{[1][8][11]} In 2018 a military version of the MDR, the NGSAR, was submitted, but was not one of the finalists in the United States Army Next Generation Squad Weapon Program trials to replace standard issue US army Infantry Weapons.^[3]

The MDR's first public debut was in 2014 at Shot Show.^{[11][2]}

Design

The Desert Tech MDR is a gas operated, conventional rotary bolt bullpup rifle. The gas system features a gas piston located above the barrel and a 6 position manual gas regulator.^[8]

The MDR rifle is designed to be ambidextrous.^{[7][2][12]} The MDR contains an ambidextrous forward and side eject port and viewing window that allows for tool free forward ejector side change.^[12] A non reciprocating ambidextrous charging handle is installed on both sides of the gun, and two or three-position safety / selector levers and magazine release are provided on both sides of the pistol grip.^[8] The rifle is compatible with Desert Tech caliber conversion kits that allow the rifle to change caliber.^{[11][2]} This provides a unique feature in which the rifle chassis can accept both intermediate cartridges and full-powered rifle cartridges on the same serial number platform (such as 5.56×45mm NATO and 7.62×51mm NATO magazines).

The outer shell of the gun is made from impact-resistant polymer with an aluminium receiver.^[8]

Many of the MDR rifle components are produced using modern Metal injection molding (MIM) metalworking processes that are becoming more common in the firearm industry. MDR MIM parts including the Barrel Block (Trunnion), Gas Block, Bolt Catch, 7.62 mag conversion bar, ejector retainer, and extractor link, are produced by Indo-MIM Pvt. Ltd.,

Micro Dynamic Rifle eXtreme	
	
A Compliant Desert Tech MDRx Rifle with 3-15 Scope, Bipod, front QD, Ratchet Compensator, Saddle Blanket, Front and Rear backup sights, over molded hand guard, and Trigger Lock	
Type	Bullpup Semi-automatic rifle Assault rifle Battle rifle Carbine Automatic rifle
Place of origin	United States
Production history	
Designer	<u>Desert Tech</u>
Manufacturer	Desert Tech
Unit cost	\$2099 (August 2020 for 5.56) ^[1]
Produced	■ 2017–2019 (MDR)^[2] ■ 2018 (NGSAR)^[3] ■ 2020–present (MDRX)
Variants	■ MDR^[1] ■ MDR-A (select-fire) ■ MDR-C (Compact)^{[4][5]} ■ NGSAR (select-fire)^[3] ■ MDRX ^[1] ■ MDRX-SE ^[1] ■ MDRX Compliant ^[1]

India. Indo-MIM Pvt, Ltd received the 2017 distinction in hand/tools award by PIM international for their innovation in MIM development and part cost reduction. ^[13]

The .308/7.62 MDR was 10,000 round endurance tested with .308 win 150 grain FMJ American Eagle, 7.62x51 149 grain American Eagle, 7.62 m80 ball US Surplus and averaged 1 failure in 2000 rounds. ^[14]

NGSAR Prototype, Modified MDR for NGSW

In 2018 a modified MDR family rifle, the NGSAR, chambered in a proprietary 6.8mm polymer-cased ammunition cartridge made by PCP was selected as one of the United States Army's Next Generation Squad Weapon Program (NGSW) weapons trial finalists to replace the United States Army's 5.56×45mm NATO caliber weapon systems. Desert Tech partnered with PCP tactical who provided a custom ammunition solution. The NGSAR was trialed in a carbine and automatic rifle configuration and were ultimately rejected. ^[3]

The NGSAR rifle included improvements in the design such as a T-Worx battery powered integrated data rail, onboard round counter, longer receiver, bolt carrier, barrel extension, charging handle, forward eject system, barrel mounting system, proprietary 6.8mm PCP magazines, custom flow reflex suppressor, CNOD day/night optic. ^[3]

MDR .308/7.62 upgrade program

The 2017 MDR in .308/7.62 win received public feedback that resulted in an upgrade program by Desert Tech on the gas and extractor system in 2019 to improve the rifle's ammunition tolerance without having to adjust the gas setting as frequently. ^[2] This update is considered the Generation 2 MDR Gas System. ^[12]

Desert Tech publicly addressed the MDR criticism on January 10, 2019. Desert Tech directly attributed the most vocal criticism of the platform to the InRangeTV company in collaboration with the Forgotten Weapons Company that demonstrated numerous failures in the system that were identified to be from varying ammunition loads used in the gas system on December

- MDRX-SE Compliant^[1]
- MDRX Micron (conversion kit)

Specifications

Mass

- 3.22 kg (7.1 lb) MDR-C^{[4][5]}
- 3.72 kg (8.2 lb) SE 16"
- 3.94 kg (8.7 lb) FE 16"
- 4.03 kg (8.9 lb) FE 20"
- 4.5 kg (9.9 lb) NGSAR ^[3]

Length

- 508 mm (20.0 in) MDR-C^[4]
- 698 mm (27.5 in) FE 16"
- 800 mm (31 in) FE 20"
- 806 mm (31.7 in) NGSAR ^[3]

Barrel length

- 267 mm (10.5 in) MDR-C^{[4][5]}
- 410 mm (16 in)^[2]
- 510 mm (20 in)^[6]

Width

- 57.4 mm (2.26 in) SE ^[1]
- 59.2 mm (2.33 in) FE ^[1]

Height

- 200.4 mm (7.89 in)^[1]

Cartridge

- .223 Remington^{[7][8]}
- 5.56×45mm NATO^{[7][8]}
- .223 Wylde^{[8][2]}

13, 2018.^[15] In addition to addressing the criticism, Desert Tech provided a technical root cause analysis presentation on the platform failures and resulting improvements with the generation 2 gas system. As a supplemental incentive, Desert Tech announced their MDR generation 2 gas system upgrade was to be covered under their warranty program for all owners.^[14]

MDRX redesign

The Micro Dynamic Rifle eXtreme, MDRX,^[16] variant was produced and marketed as a separate firearm in 2020. Improvements to the firearm include the following: Improved impact-resistant polymer,^[6] improved trigger,^[6] improved gas block,^[6] and a new "Ratchet" compensator.^[17]

The firearm was offered in .300 BLK, 6.5 Creedmoor (The first production semi-auto bullpup chambered in this cartridge), along with its original .223/5.56 and .308/7.62 calibers^{[1][8]} as well as 20" barrel and side ejection options.^[6] This update is considered the generation 3 MDR gas system.^[12]

Desert Tech halted production of the .300 BLK Rifle and conversion kits in late 2020 up until January of 2023.^[18]

A California compliant variant was produced with a 20" barrel, shark fin grip, reduced magazine size, and California compliant Ratchet compensator to qualify for a featureless rifle restrictions.^{[1][6]}

The MDRX, with its various caliber kits, is rated from the manufacturer between 1-2 MOA.^[19] A large multi shooter community study in 2022, including aftermarket precision vendors such as ES Tactical, was able to quantify the performance of the stock OEM barrel between 1.9 and 5.6 MOA with 5 shot groups in .308 with various ammunition as well as identified a contributing failure mode of Trunnion fasteners loosening over time.^[20]

MDR/MDRX official variant updates/upgrades

	▪ .300 BLK^{[7][8][2]} ▪ .308 Win^{[7][8][2]} ▪ 7.62×51mm NATO^{[7][8]} ▪ 6.5mm Creedmoor^[8] ▪ 6.8mm PCP^[3] ▪ .350 Legend^[9] ▪ .450 Bushmaster^[9]
Action	Short-stroke gas-operated piston, rotating bolt ^[8]
Rate of fire	▪ 600–650 rounds/min (MDR-A) ▪ 650 rounds/min (NGSAR^[3])
Muzzle velocity	▪ 3150 ft/s (960 m/s) (With 5.56x45mm NATO M855A1 ammunition)^[10] ▪ 2933 ft/s (894 m/s) with 6.8mm PCP in NGSAR^[3]
Effective firing range	900m ^[1]
Feed system	10, 20, & 30-round box magazine
Sights	Picatinny rail for mounting Picatinny mounted iron sights or various optical sights



Desert Tech MDRX 5.56/.223 Conversion Kit includes barrel, mag catch, mag insert, bolt head, and ejector chute

On April 4, 2019, Desert Tech announced multiple MDR updates to improve rifle performance on the 308 Rifle. The manufacturer indicated that these updates would be incorporated in all post-announcement shipments. These updates including the 2019 6-hole gas valve and wider extractor. In addition a heavy hammer spring is available for special order to resolve light strike failures on hardened primers.^[21]

On August 13, 2019, Desert Tech announced a forward eject panel spring buffer kit for sale to resolve forward eject panel spring weakening over time on the MDR platform.^[22]

On January 9, 2020, Desert Tech announced the MDRx Rifle was available for sale.^[23]

On January 28, 2021, the MDRX Micron conversion kit was announced that allowed the purchase of a 11.5" 5.56/.223 Wylde barrel, replacement hand guard, ejector plates, a new 'Raiden' flash hider, and bolt replacement for a side eject bullpup with a total length of 23.43".^[24] These kits allow standard MDRX and MDR rifles to be classified and regulated as a short-barreled rifle (SBR) requiring the user to apply with the appropriate regulatory bodies before install of the conversion kit in their rifle. These kits became available for sale on January 29, 2021.^[25]

On March 18, 2021, a collaboration between Desert Tech and BLK LBL Bipods was announced to develop, produce, and sell the Mantis handguard. The handguard is a 9-inch-long (230 mm), 16.7-ounce (470 g), aluminium handguard with an integrated bipod that replaces the factory polymer handguard and improved zero for handguard mounted parts.^[26] The aluminium handguard with integrated bipod is designed for both 16" and 20" barrel MDR/MDRX rifles and allows both an M-LOK under handguard attachment as well as an integrated bipod to be used at the same time without interference.

In early 2022, a 13-inch-long (330 mm), 20.7-ounce (590 g) Mantis handguard for the longer 20" barrel option was available for sale.^[26]

In March 2022, ES Tactical, a Desert Tech Dealer who specializes in precision upgrade parts,^[27] began to sell after market caliber conversion kits in .350 Legend and .450 Bushmaster, resulting in the first bullpup to be chambered in those cartridges.^[9]

In June 2022, Desert Tech and BLK LBL announced a 13-inch-long (330 mm) and 9-inch-long (230 mm) ALX hand guard solution similar to the Mantis without the bipod attachment.^[28]

In January 2023, Desert tech announced multiple MDRx Rifle updates to resolve rifle weaknesses. Desert Tech indicated that these updates would be incorporated in all post announcement MDRx shipments. These updates include a new magazine transfer bar, magazine catch, updated non folding charging handle system (charging handle post, charging handle, and recoil buffer), a press-fit gas block, and barrel gas port inside bevel.^[29]

In August 2023 Desert Tech announced that the picatinny rail mounted gas blocks in the barrel assembly, designed to operate with a Desert Tech MDR red dot mount, would be discontinued. All future MDRx Rifles and Barrel Assemblies would no longer include a Pic rail mounted gas block, be lighter, and allow



Desert Tech MDR (Top) and MDRX (Bottom) with various conversion kits, suppressors, various muzzle devices, magazines, necessary conversion tools, various handguards, and custom caliber ES Tactical barrels

for removal of the barrel assembly without removal of the handguard. In addition, the company distributed an email notice to its customers, on August 18th, indicating that a registered gunsmith could remove the pic rail from any existing barrel assembly without voiding the factory lifetime warranty. ^[30]



2023 MDRx charging handle

See also

Forward-ejection rifles

- [FN F2000](#)
- [A-91](#)
- [Kel-Tec RFB](#)

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